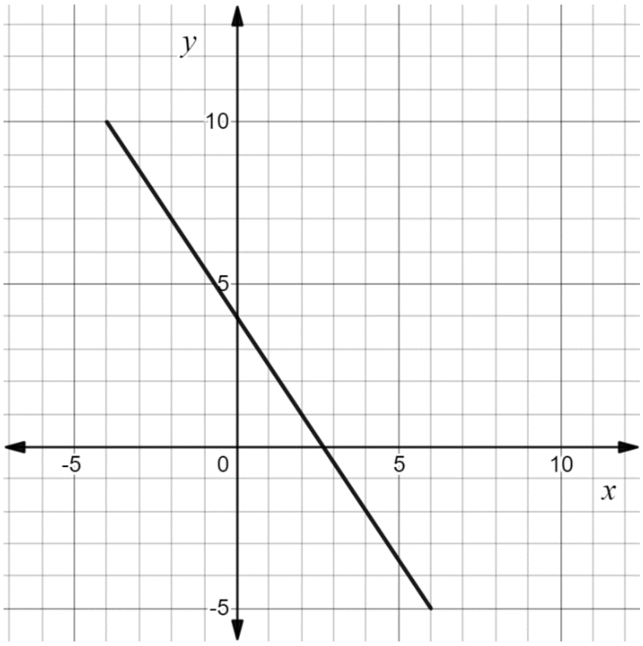
Algebra 1
Unit 4
Lesson 1 Homework

Name Answer Key____________
Date____________________
Period___________________

The graph of $y = -\frac{3}{2}x + 4$ is shown. The graph has a restricted domain.



1. Give a written description for the domain.
   All values greater than or equal to -4 and less than or equal to 6.

2. Write the range using inequalities.
   $-5 \leq y \leq 10$

3. Write the domain using set-builder notation.
   $\{x: -4 \leq x \leq 6\}$

4. Write the range using set-builder notation.
   $\{y: -5 \leq y \leq 10\}$

For each written description complete the table.

Written Description	Inequalities	Set-builder notation
5. All values less than or equal to -4	$x \leq -4$	$\{x: x \leq -4\}$
6. All real numbers	$-\infty < x < \infty$	$\{x: x \in \mathbb{R}\}$
7. All values greater than $-\frac{11}{3}$ but less than or equal to $\frac{17}{3}$	$-\frac{11}{3} < x \leq \frac{17}{3}$	$\left\{x: -\frac{11}{3} < x \leq \frac{17}{3}\right\}$

8. Write the set of values represented by $-6.7 \leq x < 9.1$ using set-builder notation.
$\{x: -6.7 \leq x \leq 9.1\}$

9. Write a written description of the set of values represented by $\{y|\ y \geq -0.25\}$.
All values greater than or equal to -0.25.

10. Write the set of values represented by $-14.25 \leq b \leq -8.37$ using set-builder notation.
$\{b: -14.25 \leq b \leq -8.37\}$